

Coupling the LES with CLEM

Let's define as U as the conservative vector such as $U = [\rho, \rho \mathbf{u}, \rho e, \rho E, \rho \mathbf{Y}]$. In the following model, the LES strategy updates exclusively $\rho, \rho \mathbf{u}, \rho E$, where $E = e + \frac{1}{2}|\mathbf{u}|^2$.

1 Notation

- c : LES cell label; $\mathcal{C}_c$ denotes the cell (its integer coordinates on the AMR level).
- $p = 1, \dots, N_p$: particle (LEM element) index within a cell.
- $k = 1, \dots, N_s$: species index; j is the dummy index in sums over species.
- i : spatial direction index (Einstein summation), as in $u_i \partial / \partial x_i$.
- $\mathbf{u} = (u_1, u_2, u_3)$: velocity vector; $\mathbf{Y} = (Y_1, \dots, Y_{N_s})$: species mass fractions; X_k : mole fraction of species k .
- P : pressure (capitalized to avoid a clash with the particle index p).
- $\widetilde{(\cdot)}$: resolved (Favre-filtered) quantity; $(\cdot)'$: sub-grid fluctuation.
- ζ : physical coordinate along the 1-D LEM domain of a cell, $\zeta \in [0, L]$; m : mass (Lagrangian) coordinate along that domain; A : constant cross-sectional area of the 1-D domain.
- λ : thermal conductivity. The LEM eddy event-rate parameter, when introduced, will be denoted Λ to avoid a clash.
- Ψ_{LES} : super-grid (LES) time-integration increment; $\mathcal{E}_c$ is reserved for the particle ensemble of cell c .

2 CLEM Definition

The sub-grid model, CLEM, solver the scalar transport ρY_k . Accordingly, there must be a coupling strategy to connect both. Note that CLEM is a combustion model that solve the zero Mach one dimensional problem in each LES cell. Then, let's define a set of particles, ensemble, in each LES cell

$$\mathcal{E}_c = \{\mathcal{P}_p^{(c)}\}_{p=1}^{N_p}$$

A particle p in a cell $\mathcal{C}_c$ is denoted as $\mathcal{P}_p^{(c)}$, and Each particle contains the following information

$$\mathcal{P}_p^{(c)} = [T_p, \mathbf{Y}_p, m_p, v_p, P_p]$$

These correspond to temperature, the vector of species mass fractions $\mathbf{Y}_p = (Y_{p,1}, \dots, Y_{p,N_s})$, mass, volume and pressure. Density is implicit given by mass and volume.

Each CLEM line is an ensemble of Lagrangian particles tied to a grid cell by position and ownership.

3 LES Time integration

The present implementation targets exclusively PeleC's Method of Lines path (`do_mol = 1`) with a single iteration (`mol_iters = 1`); the Godunov path and the SDC-type iteration are out of scope. First the LES cell is computed following a second order Method of Lines integration (Heun integration):

$$S^n = AD(U^n)U^\star = U^n + \Delta t(S^n + I_R)S^\star = AD(U^\star)U^{\star\star} = \frac{1}{2}(U^n + U^\star) + \frac{\Delta t}{2}(S^\star + I_R)$$

Here I_R is the effective reaction source. Two facts, verified against the implementation, are important:

1. **I_R is lagged during the Heun stages.** The value used in $U^\star$ and $U^{\star\star}$ is not evaluated at U^n ; it is the estimate stored in `Reactions_Type` at the end of the previous time step. (PeleC can iterate the corrector with an updated I_R when `mol_iters > 1`, an SDC-type iteration, but with `mol_iters = 1` this never occurs and the lag is always one full time step.)
2. **Layout (verified in `React.cpp`).** I_R has $N_s + 2$ components,

$$I_R = [\rho \dot{Y}_k \ (k = 1, \dots, N_s), \ \rho \dot{E}, \ \dot{Q}], \quad \dot{Q} = - \sum_{k=1}^{N_s} h_k \rho \dot{Y}_k$$

where the heat release $\dot{Q}$ is stored for diagnostics only — it is never added to the state. Moreover, I_R contributes only to the species and total-energy components; ρ and $\rho \mathbf{u}$ receive no reaction source.

At the end of `Advance.cpp`, once $U^{\star\star}$ is computed, the non-reacting source is calculated as

$$F_{AD} = \frac{1}{\Delta t}(U^{\star\star} - U^n) - I_R = \frac{1}{2}(S^n + S^\star)$$

where the second equality follows from the Heun stages (the lagged I_R cancels exactly), so F_{AD} is the time-centered non-reacting source.

In the standard PeleC path (no CLEM), the reaction update is then performed by `react_state()`, which calls `ReactorBase::react()`. It is **not** an explicit addition of a precomputed I_R : starting from the initial condition U^n , the reactor integrates the stiff ODE system over $[t^n, t^n + \Delta t]$ with F_{AD} held frozen,

$$\frac{d(\rho Y_k)}{dt} = F_{AD,k} + \dot{\omega}_k, \quad \frac{d(\rho e)}{dt} = F_{AD,e}$$

where $F_{AD,e}$ is the effective internal-energy forcing, derived from the total-energy component of F_{AD} by removing the kinetic-energy change produced by the momentum forcing. The reaction

sub-step conserves ρe (chemical energy is redistributed into sensible energy), so ρe evolves by the forcing alone while T and Y_k change. After the integration the new state is assembled as $\rho^{n+1} = \sum_k (\rho Y_k)^{n+1}$, $\rho \mathbf{u}^{n+1} = \rho \mathbf{u}^n + \Delta t F_{AD, \rho \mathbf{u}}$, and ρE^{n+1} is rebuilt from ρe^{n+1} plus the new kinetic energy. Finally, I_R is re-diagnosed as the **effective, time-averaged** reaction rate over the step:

$$I_{R,k} = \frac{(\rho Y_k)^{n+1} - (\rho Y_k)^n}{\Delta t} - F_{AD,k}, \quad I_{R,E} = \frac{(\rho E)^{n+1} - (\rho E)^n}{\Delta t} - F_{AD,E}$$

Consequently,

$$U^{n+1} = U^n + \Delta t (F_{AD} + I_R)$$

holds as an identity by construction: the actual update is the stiff ODE integration. This defines the coupling contract for the sub-grid model: CLEM must supply not an instantaneous rate $\dot{\omega}(U^n)$, but the effective reaction rate over the LES step.

This process is denoted as

$$U^{n+1} = U^n + \Psi_{LES}(\Delta t, U^n)$$

where Ψ_{LES} is the super-grid integrator.

3.1 Modifications when CLEM is active

When the sub-grid model owns combustion, the loop above changes as follows (`Advance.cpp`):

- $I_R \equiv 0$ in both Heun stages, and `react_state()` is never called: the reaction contribution must be supplied by CLEM. In particular, the reacting contribution to ρE at the super-grid must come from the CLEM feedback (open point, addressed in the coupling strategy). (This we have to study!!!!!!!!!!)
- With `clem.freeze_species = 1`, the species components of the MOL source are zeroed in both stages, so the super-grid does not transport ρY_k ; the sub-grid owns species transport entirely.
- The face mass fluxes of both MOL stages are captured and averaged, and handed to the sub-grid for the splicing step (large-scale advection of the particle ensembles). Using the same fluxes that advance the resolved continuity equation keeps the sub-grid mass transport consistent with the resolved ρ update.

4 CLEM Governing equations

In the CLEM implementation, at the super-grid, the mass species are not transported. Instead, the following equation is solved using a 1-D dimension approach. Let's define the transport of any scalar ϕ as

$$\frac{\partial \phi}{\partial t} + u_i \frac{\partial \phi}{\partial x_i} = \frac{\partial}{\partial x_i} \left(\rho D_\phi \frac{\partial \phi}{\partial x_i} \right) + \frac{\dot{\omega}_\phi}{\rho}$$

where $u_i \frac{\partial \phi}{\partial x_i}$ is the advection due to small and large scales. Note that the density is kept inside the diffusion derivative: pulling ρ out of $\partial/\partial x_i$ is valid only for constant density, which does not hold across a flame. Then, it is possible to split $u_i = \tilde{u}_i + u'_i$, where $\tilde{u}_i$ and u'_i stand for the resolved, large-filtered, and sub-grid, small, advection. Then, the equation can be regrouped as,

$$\frac{\partial \phi}{\partial t} + \tilde{u}_i \frac{\partial \phi}{\partial x_i} = 0$$

$$\frac{\partial \phi}{\partial t} + u'_i \frac{\partial \phi}{\partial x_i} = \frac{\partial}{\rho \partial x_i} (\rho D_\phi \frac{\partial \phi}{\partial x_i}) + \frac{\dot{\omega}_\phi}{\rho}$$

Note that the transport equation of a scalar is split into large and small advection; both equations advance the same field ϕ , each one being a fractional step of the split. Above, in given in 3-D form, which is intractable in complex flow configuration. Then, the small-scale ϕ transport is solved in a 1-D domain within a LES cell $\mathcal{C}_c$ leading to the non-conservative form:

$$\frac{\partial \phi}{\partial t} + F_{stirr} = \frac{\partial}{\rho \partial \zeta} (\rho D_\phi \frac{\partial \phi}{\partial \zeta}) + \frac{\dot{\omega}_\phi}{\rho}$$

where $\zeta \in [0, L]$ is the physical coordinate along the 1-D LEM domain of the cell (L is the domain length, of the order of the LES cell size), and F_{stirr} is the stochastic rearrangement due to small-scale advection (triplet map).

5 Mass conservation equation

From Poinson and Veynante, the general equation to transport the mass species is

$$\frac{\partial}{\partial t} \rho Y_k + \frac{\partial}{\partial \zeta} \rho u Y_k + \frac{\partial}{\partial \zeta} \rho V^C Y_k = \frac{\partial}{\partial \zeta} (\rho D_k \frac{W_k}{\bar{W}} \frac{\partial X_k}{\partial \zeta}) + \dot{\omega}$$

where V^C is the correction velocity to satisfy the mass conservation, and is expressed as

$$V^C = \sum_{k=1}^N D_k \frac{W_k}{\bar{W}} \frac{\partial X_k}{\partial \zeta}$$

Accordingly, the states of any particle can be described by the following set of equations:

$$\begin{aligned} \rho \frac{\partial Y_k}{\partial t} + F_{k,stirr} &= \frac{\partial}{\partial \zeta} (\hat{D}_k \frac{\partial X_k}{\partial \zeta}) - Y_k \sum_{j=1}^{N_s} \hat{D}_j \frac{\partial X_j}{\partial \zeta} + \dot{\omega}_k \\ \rho V_k Y_k &= -\hat{D}_k \frac{\partial X_k}{\partial \zeta} + Y_k \sum_{j=1}^{N_s} \hat{D}_j \frac{\partial X_j}{\partial \zeta} = -S_k \end{aligned}$$

where V_k is the diffusion velocity of species k , S_k is the corresponding diffusive mass flux, and $\hat{D}_k \equiv \rho D_k W_k / \bar{W}$, with D_k the mixture-averaged diffusion coefficient, W_k the molar mass of species

k and $\bar{W}$ the mean molar mass. This is the same mixture-averaged flux (with correction velocity, so that $\sum_k S_k = 0$ since $\sum_k Y_k = 1$) used by PeleC/PelePhysics at the super-grid.

$$\rho c_p \frac{\partial T}{\partial t} + F_{T, stirr} = \frac{\partial}{\partial \zeta} \left(\lambda \frac{\partial T}{\partial \zeta} \right) + \sum_{k=1}^{N_s} c_{p,k} S_k \frac{\partial T}{\partial \zeta} - \sum_{k=1}^{N_s} h_k \dot{\omega}_k$$

This temperature form follows from the enthalpy equation $\rho Dh/Dt = \partial_\zeta (\lambda \partial_\zeta T) - \partial_\zeta \sum_k h_k (\rho Y_k V_k)$ combined with the species equation, using $h = \sum_k Y_k h_k$ and $dh_k = c_{p,k} dT$. Note that the interspecies enthalpy transport appears as the gradient product $\sum_k c_{p,k} S_k \partial_\zeta T$, and not as a flux $\sum_k h_k S_k$ inside the divergence: keeping the full enthalpy flux inside the divergence together with the $-\sum_k h_k \dot{\omega}_k$ term would double-count the contribution $\sum_k h_k \partial_\zeta S_k$, which is already accounted for through the species equation.

5.1 Mass (Lagrangian) coordinate

The above equations are solved in mass space. **Working assumption:** the cross-sectional area A is constant and uniform along the line (one value per cell, unchanged during a sub-step), so the mass coordinate is defined by

$$m(\zeta, t) = \int_0^\zeta \rho(\zeta', t) A d\zeta', \quad dm = \rho A d\zeta$$

which is valid for an arbitrary density profile (the relation $m = A\rho\zeta$ would hold only for uniform density and is not used). The total line mass $M_c = m(L, t) = \sum_p m_p$ is invariant under diffusion, stirring and reaction; it changes only through splicing. The mass-space domain $[0, M_c]$ is therefore **static during a CLEM sub-step** even though the physical length $L(t)$ is not.

Why mass coordinates. At zero Mach, heat release at (nearly) constant pressure changes ρ , and 1-D continuity, $\partial_t \rho + \partial_\zeta (\rho u_d) = 0$, then forces a dilatation velocity $u_d(\zeta, t)$ along the line. In physical coordinates every scalar equation would carry an advection term $\rho u_d \partial_\zeta \phi$ and the grid would have to follow the expanding fluid. The mass coordinate absorbs this exactly. Anchoring the line so that $u_d(0, t) = 0$ (only velocity differences matter), differentiating the definition of m and using continuity,

$$\left. \frac{\partial m}{\partial t} \right|_\zeta = A \int_0^\zeta \partial_t \rho d\zeta' = -A \int_0^\zeta \partial_{\zeta'} (\rho u_d) d\zeta' = -A \rho u_d(\zeta, t)$$

so, by the chain rule, for any scalar ϕ ,

$$\left. \frac{\partial \phi}{\partial t} \right|_m = \left. \frac{\partial \phi}{\partial t} \right|_\zeta + u_d \frac{\partial \phi}{\partial \zeta} = \frac{D\phi}{Dt}$$

The time derivative at fixed m is the material derivative: the dilatation advection term vanishes identically in mass coordinates, and no remeshing is needed when the gas expands. Each particle represents a rectangular section (wafer) of the cell with fixed mass m_p ; its width $\delta\zeta_p = m_p/(\rho_p A)$ adjusts automatically with density.

Spatial derivatives transform as $\partial/\partial\zeta = \rho A \partial/\partial m$. Note that $\frac{1}{\rho}F_{stirr}$ can be rewritten as F_{stirr} since is a operator, re-ordering process instead an explicit solution, Then, the generic scalar equation becomes

$$\left.\frac{\partial\phi}{\partial t}\right|_m + F_{stirr} = A^2 \frac{\partial}{\partial m} \left(\rho^2 D_\phi \frac{\partial\phi}{\partial m} \right) + \frac{\dot{\omega}_\phi}{\rho}$$

Applying the same transformation to the species and temperature equations gives the working set. Species:

$$\left.\frac{\partial Y_k}{\partial t}\right|_m + F_{k,stirr} = A \frac{\partial S_k}{\partial m} + \frac{\dot{\omega}_k}{\rho}, \quad S_k = \rho A \left(\hat{D}_k \frac{\partial X_k}{\partial m} - Y_k \sum_{j=1}^{N_s} \hat{D}_j \frac{\partial X_j}{\partial m} \right)$$

where S_k is the same quantity defined above ($S_k = -\rho V_k Y_k$), merely re-expressed through $\partial_\zeta = \rho A \partial_m$; the correction-velocity property $\sum_k S_k = 0$ is unchanged. Temperature:

$$\left.\frac{\partial T}{\partial t}\right|_m + F_{T,stirr} = \frac{A^2}{c_p} \frac{\partial}{\partial m} \left(\rho \lambda \frac{\partial T}{\partial m} \right) + \frac{A}{c_p} \sum_{k=1}^{N_s} c_{p,k} S_k \frac{\partial T}{\partial m} - \frac{1}{\rho c_p} \sum_{k=1}^{N_s} h_k \dot{\omega}_k$$

Thermal variable and the role of internal energy. At the CLEM level only the temperature transport above is solved: T is the thermal unknown of the line, and no separate energy PDE is carried by the sub-grid. Internal energy is a derived quantity, computed from the particle state through the EOS, $e_p = e(T_p, \mathbf{Y}_p)$, whenever an energy representation is required:

- the reaction stage (CVODE) integrates the chemistry in an energy formulation, so each particle's e_p is assembled from $(T_p, \mathbf{Y}_p)$ before the call and T_p is recovered from the integrated energy and composition afterwards;
- the communication with the super-grid is via internal energy: the CLEM feedback to the LES enters through ρe (hence ρE), never through temperature directly.

Note on the discrete realization: the diffusion operator may equivalently be advanced in the conservative energy form,

$$\left.\frac{\partial e}{\partial t}\right|_m = A \frac{\partial}{\partial m} \left(\rho \lambda A \frac{\partial T}{\partial m} + \sum_{k=1}^{N_s} h_k S_k \right)$$

with the enthalpy flux inside the divergence and T recovered from $(\rho, e, \mathbf{Y})$ afterwards; no $-\sum_k h_k \dot{\omega}_k$ term appears there because the formation enthalpies travel inside e . This form is the same equation in the continuum, and discretely it conserves internal energy on the line to round-off — it is the realization currently used by `ClemDiffusion.cpp`.

Discrete (wafer) form. In the present implementation, Δm is uniform before starting the CLEM operation, by implementing the reggridding operation. This is equivalent to ha a uniform spatial grid. To discretize, a central difference scheme in a discrete element framework is employed. In particular, the second order derivative uses an $m \pm 1/2$ strategy. That is

$$\frac{\partial}{\partial m} \left(D \frac{\partial \phi}{\partial m} \right) \approx \frac{1}{\Delta m} \left(D_{p+1/2} \frac{\partial \phi}{\partial m} \Big|_{m+1/2} - D_{p-1/2} \frac{\partial \phi}{\partial m} \Big|_{m-1/2} \right)$$

where value at $m \pm 1/2$ are evaluated using a harmonic mean $D_{m+1/2} = \frac{2D_{m+1}D_m}{D_{m+1}+D_m}$. The gradient at $m \pm 1/2$ are evaluated using the following

$$D_{p+1/2} \frac{\partial \phi}{\partial m} \Big|_{m+1/2} \approx D_{p+1/2} \left(\frac{\phi_{m+1} - \phi_m}{\Delta m} \right)$$

Throughout, the widths $\delta\zeta_p = m_p/(\rho_p A)$ adjust with density, so the line length

$$L(t) = \sum_p \frac{m_p}{\rho_p A}$$

expands or contracts with heat release and compression while the mass grid $\{m_{p+1/2}\}$ never moves during a sub-step. This $L(t)$ — equivalently the ensemble volume $\sum_p m_p/\rho_p$ — is the natural carrier of the sub-grid dilatation information that must be returned to the super-grid (the ρE feedback flagged as an open point above).

Finally, the triplet map is naturally formulated in mass space: a stirring event is a measure-preserving rearrangement of a sub-interval of $[0, M_c]$, so it permutes mass rather than length and leaves the fixed mass grid untouched; with equal-mass particles it reduces to an index permutation of the ensemble.

The equations are advanced by operator splitting. Three operators act on the ensemble state of a cell:

- $\mathcal{O}_{diff}(\delta t)$: advances the diffusion terms (the flux divergences in the species and temperature equations);
- $\mathcal{O}_{stirr}$: applies the triplet-map rearrangements associated with F_{stirr} ; it is not a continuous-in-time operator but a sequence of instantaneous stirring events at stochastically sampled times;
- $\mathcal{O}_{react}(\delta t)$: advances the chemical source terms ($\dot{\omega}_k$ and $-\sum_k h_k \dot{\omega}_k$).

Diffusion and reaction march together in small sub-steps, while $\mathcal{O}_{stirr}$ fires as instantaneous events at stochastically sampled times. The sub-grid equations are advanced over the full super-grid increment Δt_{LES} by the following sub-cycling loop (per cell; $t_\Delta = t^n + \Delta t_{LES}$ is the end of the LES step and t is the sub-grid clock):

1. Sample the time to the next stirring event, Δt_{stir} , from the eddy event rate, and set $t_{stir} = t + \Delta t_{stir}$. (The event rate and eddy-size distribution closing $\mathcal{O}_{stirr}$ from the LES sub-grid state — parameter Λ — remain to be specified.)
2. Compute the diffusion sub-step Δt_{diff} from the explicit-diffusion stability (CFL) condition on the line: $\Delta t_{diff} \leq C_{diff} \Delta m^2 / [A^2 \max_{p,k} (\rho_p^2 D_{k,p}, \rho_p \lambda_p / c_{v,p})]$.
3. Clip the sub-step so it overshoots neither the end of the LES step nor the next stirring event: $\Delta t_{diff} \leftarrow \min(\Delta t_{diff}, t_\Delta - t, t_{stir} - t)$.

4. **Diffusion step** $\mathcal{O}_{diff}(\Delta t_{diff})$: advance the species and temperature equations on the Δm line.
5. **Reaction step** $\mathcal{O}_{react}(\Delta t_{diff})$: advance the chemical sources over the same sub-step.
6. $t \leftarrow t + \Delta t_{diff}$. If $t \geq t_{stir}$: apply the triplet-map stirring event $\mathcal{O}_{stirr}$ and resample $t_{stir} = t + \Delta t_{stir}$.
7. If $t < t_{\Delta}$, repeat from step 2.

The stability limit of step 2, written in mass units, reduces via $\Delta m = \rho A \delta \zeta_p$ to the familiar physical-space bound $\sim \delta \zeta_p^2 / (2D)$, as it must. Two implementation notes (`ClemDiffusion.cpp::stableSubStep`): the species coefficient is $\rho^2 D_k = \rho \hat{D}_k \bar{W} / W_k$, **not** $\rho \hat{D}_k$ — the transported variable is Y_k while the driving gradient is in X_k , and the factor $\bar{W} / W_k$ (~ 27 for H in air) is what keeps the light radicals inside their stability limit; and the thermal coefficient carries c_v , not c_p , because the energy update goes through e .

Each pass of the loop is the first-order (Lie) split $\mathcal{O}_{react}(\Delta t_{diff}) \circ \mathcal{O}_{diff}(\Delta t_{diff})$, with stirring interleaved between passes whenever the sub-grid clock crosses a sampled event time. The CLEM clock is therefore constrained by the diffusion stability and by the stirring event sequence, never by the super-grid step: both Δt_{diff} and Δt_{stir} are in general much smaller than Δt_{LES} . In the current implementation the diffusion stage already runs exactly this sub-cycle over the whole Δt_{LES} (`clem.diffusion_cfl`, capped by `clem.diffusion_max_substeps`); stirring and reaction are upcoming and will interleave into the same loop, matching the stage ordering of `ClemAlgorithm.cpp` (isentropic pressure update, regrid, diffusion — then stirring and reaction — then splicing for the inter-cell transport at the LES level).

6 Consideration

The Zero-Mach equation are solved using a Δm equal for all particles.

1. All particle adopt the pressure from the LES cell, $\tilde{p}$. The other properties, such as temperature, are updated using the isentropic relations.
2. The regridding peprator $\mathcal{O}_{regrid}$ is applied in each LEM domain such as Δm is uniform in every particle within the LEM domain
3. The zero-Mach equation are solved in a smaller time step, Δt_{CLEM} such as the stabiltiy of the diffusion and the stirring are full filled.

7 CLEM Communication to LES

The coupling-back operator returns the sub-grid's effect on to the super-grid state $U(\rho \mathbf{Y}, \rho e)$. Two ensemble averages of a particle field ϕ_p over $\mathcal{E}_c$ are used: the Favre (mass-weighted) mean, and its conserved (density) form,

$$\bar{\phi}_c = \frac{\sum_{p \in \mathcal{E}_c} m_p \phi_p}{\sum_{p \in \mathcal{E}_c} m_p}, \quad [\rho \phi]_c = \frac{\sum_{p \in \mathcal{E}_c} m_p \phi_p}{V_c}$$

with $V_c = \Delta_x^{DIM}$ the LES cell volume. Two couple back the CLEM, the following strategy is designed enforcing that the CLEM is the only mechanism responsible for the scalar transport,

i.e, the $\bar{\rho}\tilde{Y}_k$, That is, during the RK2 integration, prediction and correction, the scalar are frozen. CLEM owns $\rho\mathbf{Y}$ transport entirely. Then, the scalar update and coupling is defined as follows:

$$\Delta[\rho Y_k]_c = [\rho Y_k]_c^{post-splice} - [\rho Y_k]_c^{pre}, \quad (\rho Y_k)_c^{n+1} = (\rho Y_k)_c^n + \Delta[\rho Y_k]_c$$

then **projected** onto the LES density, $(\rho Y_k)_c^{n+1} \leftarrow (\rho Y_k)_c^{n+1} \rho_c^{n+1} / \sum_j (\rho Y_j)_c^{n+1}$, so the denisty solved in the LES cell, $\sum_k \rho Y_k = \rho_{LES}$, holds exactly regardless of splice/regrid round-off. The energy channel is treated the same way, $\Delta[\rho e]_c$ added to $(\rho e)_c^n$, with ρE^{n+1} rebuilt from the LES kinetic energy. The energy update is required because of the volumen normalization, $\mathcal{O}_{vol}$ (which isentropically rescales the line back to V_c), so the line's $p dV$ breathing work is captured in $\Delta[\rho e]_c$ instead of being lost to the fixed-volume LES cell.

This strategy leaves ρ and $\rho\mathbf{u}$ untouched, only $\rho\mathbf{Y}$ and, ρe are written back. This is consistent with the LES formualtion, owning mass and momentum exclusively.

The volume renormalization $\mathcal{O}_{vol}$ si required due to the isobaric coupling, which leads to a cell colume drift after the CLEM operations. Accordignly, immediately before the conserved snapshot, every particle is rescaled by the same isentropic ratio $f = V_c / \sum_{p \in \mathcal{E}_c} v_p$:

$$\rho_p \rightarrow \rho_p / f, \quad T_p \rightarrow T_p f^{-(\gamma_p - 1)}, \quad P_p \rightarrow P_p f^{-\gamma_p}, \quad e_p \leftarrow e(\rho_p / f, T_p f^{-(\gamma_p - 1)}, \mathbf{Y}_p) - e(\rho_p, T_p, \mathbf{Y}_p)$$

so that $\sum_p v_p \rightarrow V_c$ exactly. This turns the line's $p dV$ breathing work back into internal energy rather than losing it to the LES's fixed-volume cell, and is what the energy channel of the conserved-delta write-back above ultimately captures.

Overall, the strategy summarizes as follows,

$$U^{n+1,\star} = U^n + \Delta t \Psi_{LES}$$

where the mass species are kept frozen, and only denisty, momemtum and total energy are udated here. Later, we apply the CLEM strategy to account for the reaction and the transport of scalars

$$U^{n+1} = U^{n+1,\star} + \Delta t \Psi_{CLEM}$$

Here the compressibility effect, reaction and scalar trasnport are accouter for following the strategy described above.